YASAFlaskified & psgscoring

An Open-Source Platform and Python Library for Automated Polysomnography Analysis Using AI-Based Sleep Staging and AASM 2.6–Compliant Respiratory Scoring

Bart Rombaut, MD[1], *[Open position for contributors]*[2], and Raphael Vallat, PhD[3]

[1]Department of Pneumology, Slaapkliniek AZORG, Aalst, Belgium
[2]*[Reserved for YASA developers / clinical collaborators]*
[3]Center for Human Sleep Science, University of California, Berkeley, CA, USA

March 2026

Corresponding author: Bart Rombaut MD, Department of Pneumology, AZORG, Merestraat 80, 9300 Aalst, Belgium. Web: `www.slaapkliniek.be`

Abstract

We present YASAFlaskified (v0.8.1) and `psgscoring`, a companion open-source Python library for AASM 2.6–compliant respiratory event scoring. The system extends YASA—developed by Vallat and Walker at UC Berkeley [2]—into a complete clinical PSG analysis platform. `psgscoring` implements validated signal processing algorithms requiring no deep learning or GPU: square-root linearization of nasal pressure signals [12, 13], Mean Magnitude of Second Derivative (MMSD) apnea validation [14], dual-sensor detection with exclusion masking, temporally constrained SpO_2 coupling [16], and a novel two-pass Rule 1B pipeline with breath-cycle validation. Version 0.8.1 introduces rolling 2-minute arousal baselines adapted to fragmented sleep, ratio-based sigma-band spindle exclusion, clipping-to-artifact feedback for uncontaminated TST calculation, FHIR R4 treatment recommendations, multi-site access control with role-based data isolation, and an interactive EDF browser with channel-group filters. The platform is deployed as a Docker Compose application. `psgscoring` is distributed under the BSD-3 license (`pip install psgscoring`).

Keywords: polysomnography, sleep staging, YASA, AASM 2.6, obstructive sleep apnea, AHI, respiratory scoring, signal processing, nasal pressure linearization, Python library, psgscoring

1. Introduction

Polysomnography (PSG) remains the gold standard for diagnosing sleep-disordered breathing, periodic limb movement disorder, and other sleep pathologies [1]. Manual scoring requires 2–4 hours per study and is subject to 80–90% inter-scorer agreement [5], creating bottlenecks in sleep medicine.

YASA (Yet Another Spindle Algorithm), developed by Vallat and Walker at UC Berkeley [2], provides a validated LightGBM-based sleep staging classifier achieving ∼85% epoch agreement. However, YASA focuses on EEG-based staging and does not address respiratory scoring, arousal detection, PLM analysis, or clinical reporting.

For respiratory event detection, **no established Python library exists** implementing full AASM 2.6–compliant scoring. Published algorithms have been validated individually:

- Monserrat/Thurnheer [13, 12]: $\sqrt{\cdot}$-linearization of nasal pressure (r^2=0.88–0.96)
- Lee et al. [14]: MMSD method (92% agreement, κ=0.78)
- Nakano et al. [15]: Rule-based nasal pressure (86.4% sensitivity)
- Uddin et al. [16]: Airflow + SpO_2 coupling

YASAFlaskified integrates these validated approaches into a unified clinical platform, and `psgscoring` extracts the core algorithms into a standalone, pip-installable Python library (BSD-3 license) for the research community.

2. The YASA Library

2.1 Origin and Development

YASA was created by Raphael Vallat at the Center for Human Sleep Science directed by Matthew Walker at UC Berkeley. First released in 2018, it evolved into a comprehensive sleep analysis library published in *eLife* [2]. Vallat is also the author of Pingouin [10]. YASA is built on NumPy, SciPy, MNE-Python [8], and scikit-learn, distributed under the BSD-3 license with over 500 citations.

2.2 LightGBM Sleep Staging

The core classifier uses LightGBM [9], trained on SHHS and MESA data. Per 30-second epoch, features are extracted from three channels:

EEG: Spectral power in six bands (δ, θ, α, σ, β, γ) via Welch's method; spectral entropy; Hjorth parameters; zero-crossing rate; permutation entropy.

EOG: Low-frequency power (<2 Hz) for slow eye movements; rapid eye movement quantification for REM identification.

EMG: RMS amplitude of chin/submentalis. Optional but improves REM/Wake discrimination by 3–5%.

Temporal smoothing reduces physiologically implausible transitions. Per-epoch confidence scores enable targeted review of low-confidence epochs (<70%), reducing review effort by 80–90%.

2.3 Additional YASA Modules

Spindle detection (12–15 Hz σ band), slow-wave detection (0.3–1.5 Hz, >75 μV), sleep cycle analysis (modified Feinberg & Floyd [4]), and spectral band power computation per epoch and stage.

3. System Architecture

3.1 Technology Stack

Flask/Gunicorn (Python 3.11), Redis 7 + RQ for async jobs, MNE-Python [8] for EDF I/O, SciPy/NumPy for signal processing, YASA 0.7 [2] with LightGBM [9], ReportLab for PDF, edfio/pyedflib for EDF+ (three-tier fallback), Docker Compose with three containers (app, worker, Redis).

3.2 Processing Pipeline

Table 1: Pipeline stages (8h PSG, 256 Hz, 39 channels)

Step	Description	Duration
1. EDF loading	3-pass selective channels	13 s
2. Sleep staging	YASA LightGBM (EEG+EOG+EMG)	7 s
3. EEG micro	Spindles, SW, REM, bandpower, cycles	40 s
4. Respiratory	AASM 2.6 apnea/hypopnea (Rule 1A)	70 s
5. Ancillary	SpO_2, HR, position, snoring, PLM	20 s
6. Arousal	Two-phase detection + Rule 1B	50 s
7. Quality	Signal quality + clipping feedback	5 s
8. CSR	Cheyne-Stokes via autocorrelation	5 s
9. Reports	PDF, PSG, Excel, EDF+, FHIR R4	30 s

3.3 Three-Pass EDF Loading

Pass 1: EEG+EOG+EMG (3 channels) for staging. **Pass 2:** All EEG for spindle/SW/bandpower. **Pass 3:** Respiratory channels + primary EEG for arousals. This minimizes memory (>2 GB avoided).

3.4 Automatic Unit Conversion

MNE's `raw.get_data()` returns Volt. AASM thresholds are in μV. If $\max(|x|) < 0.01$ V, the system converts $\times 10^6$. Applied independently in arousal, PLM, and EMG processing.

3.5 Clipping-to-Artifact Feedback (v0.8.1)

Signal quality assessment detects clipping (amplifier saturation) per channel. In v0.8.1, epochs with >5% clipped samples on the primary EEG channel are added to the artifact mask *before* respiratory analysis, ensuring that saturated segments do not contaminate TST or inflate AHI.

4. The `psgscoring` Library

4.1 Motivation

While YASAFlaskified provides a complete web platform, many researchers need only the respiratory algorithms. YASA itself acknowledges this gap: *"future developments should prioritize automated scoring of clinical disorders, particularly apnea-hypopnea events"* [2]. `psgscoring` fills this gap as a standalone library (BSD-3, `https://github.com/bartromb/psgscoring`).

4.2 Architecture

Requires only `numpy` ≥1.22, `scipy` ≥1.8, and `mne` ≥1.0. Two core modules: `_core.py` (2401 lines, 44 functions) and `_arousal.py` (891 lines, 15 functions). The public API exposes 30+ functions:

```
from psgscoring import run_full_analysis
results = run_full_analysis(raw, hypno)
ahi = results['respiratory']
            ['summary']['ahi_total']
```

4.3 Validation

21 unit tests covering linearization, MMSD, flow envelope, baseline, breath detection, flattening index, and utilities. GitHub Actions CI across Python 3.9–3.12.

4.4 Current Limitations (v0.1.0)

1. **Monolithic modules.** Two large files rather than domain-specific modules.
2. **Type annotations.** Python 3.10 union syntax limits 3.9 compatibility.
3. **Logging.** Flask-style loggers; should be caller-configured.
4. **MNE coupling.** Full pipeline requires MNE `Raw` object; individual functions accept plain arrays.
5. **No benchmark validation.** Thresholds empirically tuned at AZORG, not validated on SHHS/PhysioNet.
6. **Limited documentation.** Docstrings present but no signal flow diagrams.
7. **No streaming.** All functions operate on complete in-memory arrays.

4.5 Roadmap

Phase 1 (v0.1–0.3): Modular split, Python 3.9 compat, logging fix, >80% test coverage.

Phase 2 (v0.4–0.9): Validation on SHHS/MESA/PhysioNet with published Bland-Altman and κ statistics. MNE decoupling. Sphinx documentation.

Phase 3 (v1.0): Stable API, property-based testing, Zenodo DOI, PyPI release. YASAFlaskified v1.0 would then import `psgscoring` as a dependency rather than bundling the algorithms.

5. Respiratory Event Detection

5.1 Nasal Pressure Square-Root Linearization

Nasal pressure $\propto$ flow^2 (Bernoulli). A 50% flow reduction → 75% amplitude reduction → systematic hypopnea overestimation. Following AASM Rule 3 [17]:

$$x_{\text{lin}}(t) = \text{sign}\big(x(t)\big) \cdot \sqrt{|x(t)|} \tag{1}$$

Applied exclusively to the nasal pressure (hypopnea) channel, not the thermistor (apnea). Thurnheer et al. [12] confirmed $r^2 = 0.88$–0.96 vs. pneumotachography.

5.2 Flow Envelope Computation

Band-pass filter (0.05–3 Hz, 3rd-order Butterworth), Hilbert transform for instantaneous amplitude, 1-second moving average smoothing. The 3 Hz upper cutoff implicitly removes snoring vibrations (50–200 Hz), ensuring reliable flattening index computation downstream [18].

5.3 Dynamic and Stage-Specific Baseline

Per-sample dynamic baseline via 5-minute sliding window (95th percentile), sampled every 10 s with linear interpolation (∼2500× speedup). Combined with stage-specific baselines (NREM/REM, 5 s ramp transitions) and position-reset after body movement.

5.4 MMSD Apnea Validation

The Mean Magnitude of Second Derivative [14] provides a drift-independent measure:

$$\text{MMSD}(t) = \frac{1}{N} \sum_{i=t-N/2}^{t+N/2} |x''(i)| \tag{2}$$

where N is a 1-second window. Normalized MMSD > 40% of baseline during a candidate apnea → false positive rejected. Lee et al. reported 92% agreement (κ=0.78) on 24 PSG recordings.

5.5 Dual-Sensor Detection (AASM 2.6)

Table 2: AASM 2.6 dual-sensor paradigm

Sensor	Signal	Role	Threshold
Thermistor	Thermal airflow	Apnea	<10%, ≥10 s
Nasal pressure	$\sqrt{}$-linearized	Hypopnea	10–70%, ≥10 s

Exclusion mask: ±5 s margin around each apnea prevents flank double-counting.

5.6 Apnea Type Classification

Four-step effort analysis on thoracic/abdominal RIP: (1) amplitude ratio vs. baseline (>40% obstructive, <20% central); (2) coefficient of variation; (3) cross-correlation thorax–abdomen; (4) first/second half comparison for mixed events. Composite confidence score (0–1) per event.

5.7 Breath-by-Breath Analysis

Zero-crossing segmentation (1–15 s per breath). Peak-to-trough amplitude, local baseline from median of 10 preceding breaths, and **flattening index**: fraction of inspiratory segment >80% of peak. Values >0.3 indicate flow limitation (RERA). The input signal is already bandpass-filtered at 0.05–3 Hz, removing snoring artifact before flattening computation [18].

5.8 SpO_2 Coupling and Rule 1B

5.8.1 Rule 1A (Desaturation)

Baseline: 90th percentile of 120 s pre-event. Nadir window: onset → 30 s post-event [16]. Nadir ≥3 s after onset (circulatory delay). Early nadirs with <5% desaturation rejected.

5.8.2 Rule 1B — Two-Pass with Breath Validation

Pass 1: Events without ≥3% desaturation stored as *rejected candidates*. **Pass 2:** After arousal detection, candidates re-evaluated. An arousal within 15 s of event termination triggers reinstatement as Rule 1B hypopnea.

v0.8.1 improvement: If >1 complete breath cycle occurs between event termination and arousal onset, the coupling is considered coincidental and the candidate is not reinstated. This prevents false associations where an arousal happens to fall within 15 s of a flow reduction that had already resolved with normal breathing.

5.9 Respiratory Indices

Table 3: Indices (artifact-corrected TST)

Index	Definition	Thresholds
AHI	(Apneas+hypopneas)/TST h	<5 / 5–15 / 15–30 / >30
OAHI	(Obstr. apneas+hyp.)/TST h	Excl. central/mixed
RDI	AHI + RERA index	Incl. sub-threshold + arousal
ODI	Desaturations (≥3%)/TST h	<5 normal

6. Arousal Detection

6.1 Rolling Baseline (v0.8.1)

Previous versions used a single global baseline per sleep stage (median of the quietest 50% of all NREM or REM power). In patients with severely fragmented sleep (arousal index >40), the global baseline is elevated by the arousals themselves, making the 2.0× threshold harder to reach and causing under-detection.

Version 0.8.1 replaces the global baseline with a **rolling 2-minute window** over stable sleep:

1. Every 10 s, collect arousal power from the preceding 120 s of sleep-scored samples
2. Take the median of the quietest 50% (excluding arousal-contaminated segments)
3. Linearly interpolate between anchor points for a per-sample baseline
4. Fall back to global baseline if <10 s of stable sleep is available locally

This ensures that the detection threshold adapts to local sleep depth rather than being dominated by the night average.

6.2 Two-Phase Detection

Phase 1: Regions where α_{narrow}(8–11 Hz) + θ(4–8 Hz) + β(16–30 Hz) exceeds 2.0× the rolling per-stage baseline. Label contiguous segments ≥3 s.

Phase 2: Event-level validation:

Table 4: Event-level arousal validation

Check	Criterion
A. Pre-sleep	≥60% of 10 s pre-window is sleep
B. Abruptness	First 1 s power / 3 s pre-power > 1.5×
C. Spindle	Reject only if $\sigma > 2\times$ baseline **and** $\alpha+\beta < 50\%$ of σ (v0.8.1)
D. REM EMG	EMG rise > 2× baseline for ≥1 s [1]

6.3 Spindle Exclusion (v0.8.1)

During N2 sleep, arousal bursts often coincide with terminating spindles—particularly at the moment of awakening from N2. Previous versions rejected events where arousal power was less than sigma power (absolute comparison). This caused false negatives at N2→Wake transitions where both sigma and alpha/beta are simultaneously elevated.

The v0.8.1 criterion rejects an event *only* if sigma exceeds 2× the rolling baseline **and** the combined $\alpha+\beta$ power is less than 50% of sigma. This preserves arousals where alpha/beta dominate despite coincidental spindle activity.

6.4 Frequency Bands

Table 5: Arousal detection bands

Band	Range	Role
α narrow	8–11 Hz	Core arousal (excl. spindle overlap)
θ	4–8 Hz	Theta-burst arousals (elderly)
β	16–30 Hz	Fast activity (AASM ">16 Hz")
σ	12–15 Hz	Spindles — *exclusion*

6.5 RERA Detection

Flow limitation (≥10 s, flatness ratio < 0.80) without meeting apnea/hypopnea thresholds, terminated by arousal. RERAs contribute to RDI.

7. Periodic Limb Movement Analysis

AASM 2.6 criteria [1, 7]: EMG ≥8 μV above resting (auto V→ μV), 0.5–10 s, bilateral ±0.5 s, wake excluded, respiratory-associated excluded, series ≥4 with 5–90 s intervals.

8. SpO_2 and Cheyne-Stokes

8.1 SpO_2 Analysis

Desaturation ≥3% from 60 s rolling maximum (sleep-only). Nadir ≥3 s after event onset. ODI, time below 90/85/80%. Separate NREM/REM statistics.

8.2 Cheyne-Stokes Respiration

Autocorrelation of very-low-frequency flow envelope (0.005–0.05 Hz). Peak > 0.3 in 40–120 s lag → crescendo-decrescendo pattern. Clinical: NYHA III–IV association.

9. Signal Quality Assessment

Per-channel quality (amplitude range, SNR, flatline, clipping) with labels (good/moderate/poor/unusable). Confidence-based epoch flagging (<70% max probability) identifies the 10–20% of epochs most likely misclassified.

v0.8.1: Epochs where >5% of samples are clipped (amplifier saturation) on the primary EEG channel are injected into the artifact mask before respiratory and arousal analysis. This prevents saturated segments from contaminating TST and producing false-positive respiratory events.

10. Clinical Report and Diagnostic Engine

10.1 Automated Conclusions

Table 6: Diagnostic conclusions with treatment suggestions

Condition	Treatment
Normal (AHI < 5)	None
Mild OSAS (5–15)	Position therapy, MAD, hygiene
Moderate (15–30)	CPAP first-line; MAD if intolerant
Severe (>30)	CPAP urgent, ENT, O_2
PLMI ≥ 15	Ferritin, iron, dopamine agonist
SE < 85% / TST < 360 min	CBT-I first-line
CSR detected	Cardiology, echo, ASV
BMI > 28 + OSAS	Weight reduction addendum

Conclusions editable in Dutch, French, and English. Verification workflow for technician/physician sign-off.

10.2 FHIR R4 with Treatment Recommendations (v0.8.1)

The FHIR R4 DiagnosticReport now includes treatment recommendations in the `conclusion` field, derived from the diagnostic table above. This ensures that downstream EHR systems (e.g., Epic, ChipSoft) receive actionable clinical guidance alongside the quantitative indices.

10.3 Output Formats

PDF (portrait A4, 12-section AASM format), PSG report (landscape, institutional header), Excel, EDF+ annotations, FHIR R4.

11. Interactive EDF Browser

Multi-epoch zoom (30/60/150/300 s), event overlays (OA/CA/MA/H/AR/RERA), click-to-toggle editing with real-time AHI recalculation. Channel-group filters: Neuro (EEG+EOG+EMG), Pneumo (Resp+SpO_2), Cardio (ECG), Overig (snore+position). Event list sidebar with filtering and direct epoch navigation.

12. Multi-Site Access Control

Table 7: Role-based data isolation

Role	Data access	User mgmt	Sites
admin	All sites	All users/roles	Create/edit/delete
site	Own site only	Users (own site)	View own
user	Own site only	—	—

`site_id` and `owner_username` stored per analysis. 22 data routes enforce access via `_require_job_access()`. Admin user management shows site-filtered user lists for site managers.

13. Validation Framework

Staging metrics (per-stage sensitivity/specificity/PPV/NPV, Cohen's κ, confusion matrix). AHI agreement (Bland-Altman, Pearson correlation, severity concordance). Planned protocol: Phase 1 (n=50, AI vs 1 RPSGT), Phase 2 (n=200, AI vs 2 RPSGTs), Phase 3 (multi-center).

14. Limitations

- Sleep staging ∼85%; N1 sensitivity 30–50% [5]
- Arousal thresholds (2.0×, 1.5×) empirically tuned
- $\sqrt{}$-linearization assumes quadratic P–F relationship [12]
- MMSD 40% threshold empirically determined
- RERA without esophageal pressure
- Effort classification dependent on RIP quality
- No formal clinical validation yet (framework ready, Section 13)

YASAFlaskified is a **screening tool**. Not FDA-cleared or CE-marked. MDR Class IIa pathway identified.

15. Conclusion

YASAFlaskified and `psgscoring` demonstrate that combining Vallat's validated YASA sleep staging with peer-reviewed signal processing produces clinically useful PSG analysis. Version 15.01 addresses specific weaknesses identified through external review: rolling arousal baselines for fragmented sleep, ratio-based spindle exclusion preserving N2→Wake arousals, breath-cycle validation for Rule 1B coupling, and clipping-to-artifact feedback for uncontaminated indices.

Key contributions: (1) $\sqrt{}$-linearization [13, 12]; (2) MMSD validation [14]; (3) dual-sensor detection with exclusion masking; (4) temporally constrained SpO_2 coupling [16]; (5) two-pass Rule 1B with breath validation; (6) rolling-baseline arousal detection with ratio-based spindle exclusion; (7) breath-by-breath flattening index [18]; (8) Cheyne-Stokes autocorrelation; (9) clipping-to-artifact feedback; (10) FHIR R4 treatment recommendations; (11) multi-site data isolation; and (12) validation metrics framework.

We invite the sleep medicine and biomedical engineering communities to contribute to validation and development.

Acknowledgments

The authors thank Raphael Vallat for creating and maintaining YASA. The staging classifier was trained on SHHS and MESA data funded by the NHLBI. We thank the sleep technicians at Slaapkliniek AZORG for clinical feedback.

Conflict of Interest

BR developed YASAFlaskified and practices as pneumologist at AZORG. RV developed YASA and has no financial interest in YASAFlaskified. No other conflicts.

Data Availability

YASAFlaskified: `www.slaapkliniek.be`. YASA: `https://github.com/raphaelvallat/yasa`. psgscoring: `https://github.com/`

`bartromb/psgscoring`. Validation data upon trial completion, subject to ethics approval.

References

[1] Berry RB, et al. The AASM Manual for the Scoring of Sleep and Associated Events. Version 2.6. AASM; 2020.

[2] Vallat R, Walker MP. An open-source, high-performance tool for automated sleep staging. *eLife*. 2021;10:e70092. doi:10.7554/eLife.70092. (`https://github.com/raphaelvallat/yasa`)

[3] Iber C, et al. The AASM Manual for the Scoring of Sleep and Associated Events. AASM; 2007.

[4] Feinberg I, Floyd TC. Systematic trends across the night in human sleep cycles. *Psychophysiology*. 1979;16(3):283–291.

[5] Rosenberg RS, Van Hout S. The AASM inter-scorer reliability program. *J Clin Sleep Med*. 2013;9(1):81–87. doi:10.5664/jcsm.2350

[6] Guilleminault C, et al. Upper airway resistance syndrome. *Chest*. 1996;109(4):901–908.

[7] Zucconi M, et al. WASM standards for recording and scoring PLM. *Sleep Med*. 2006;7(2):175–183. doi:10.1016/j.sleep.2006.01.001

[8] Gramfort A, et al. MNE software for processing MEG and EEG data. *NeuroImage*. 2014;86:446–460. doi:10.1016/j.neuroimage.2013.10.027

[9] Ke G, et al. LightGBM: A highly efficient gradient boosting decision tree. *Advances in NeurIPS*. 2017;30.

[10] Vallat R. Pingouin: statistics in Python. *J Open Source Software*. 2018;3(31):1026.

[11] Peppard PE, et al. Increased prevalence of sleep-disordered breathing in adults. *Am J Epidemiol*. 2013;177(9):1006–1014.

[12] Thurnheer R, Xie X, Bloch KE. Accuracy of nasal cannula pressure recordings. *Am J Respir Crit Care Med*. 2001;164(10):1914–1919. doi:10.1164/ajrccm.164.10.2102104

[13] Montserrat JM, et al. Effectiveness of CPAP treatment in daytime function in sleep apnea. *Am J Respir Crit Care Med*. 2001;164(4):608–613.

[14] Lee H, et al. Detection of apneic events from single channel nasal airflow using 2nd derivative method. *Physiol Meas*. 2008;29:N37–N45. doi:10.1088/0967-3334/29/5/N01

[15] Nakano H, et al. New rule-based algorithm for real-time detecting sleep apnea and hypopnea events. *J Clin Sleep Med*. 2016;12(10):1389–1396. doi:10.5664/jcsm.6200

[16] Uddin MB, et al. A novel algorithm for automatic diagnosis of sleep apnea from airflow and oximetry signals. *Physiol Meas*. 2021;42:015001. doi:10.1088/1361-6579/abd47a

[17] Berry RB, et al. Rules for scoring respiratory events in sleep: update of the 2007 AASM Manual. *J Clin Sleep Med*. 2012;8(5):597–619. doi:10.5664/jcsm.2172

[18] Hosselet J, et al. Detection of flow limitation with a nasal cannula/pressure transducer system. *Am J Respir Crit Care Med*. 1998;157(5):1461–1467.